Final Exam Solution EE 4051

(1) a) $\vec{E} = (j5\eta)\,\vec{a}_x\, e^{-z}\, e^{-j\sqrt{2}z}$ (V/m)

The electric field has only one component → linear polarization.

b) $\alpha = 1$ Np/m $\quad \beta = \sqrt{2}$ rad/m

$$\frac{\alpha}{\beta} = \frac{1}{\sqrt{2}} \Rightarrow \frac{\sqrt{1+(\tan\theta)^2}-1}{\sqrt{1+(\tan\theta)^2}+1} = \frac{1}{2}$$

$$\Rightarrow \tan\theta = \sqrt{8} = 2.83 \quad \text{(loss tangent)}$$

c) $\frac{S}{S_0} = e^{-2\alpha} = e^{-2} = 0.135 \Rightarrow$ 13.5 % loss

(2) a) $$|\Gamma| = \left|\frac{30 - j\frac{1}{\omega C} - 75}{30 - j\frac{1}{\omega C} + 75}\right| = \left|\frac{-45 - j15.9}{105 - j15.9}\right| = 0.45$$

$$S = \frac{1+0.45}{1-0.45} = 2.63$$

b) At higher frequencies, the impedance of the capacitor diminishes, and the magnitude of reflection coefficient decreases.

(3) Judging by the cavity dimensions, the dominant mode is TM_{110} and the next mode is TE_{101}.

The resonance frequency of TE_{101} should be equal or greater than 5 GHz.

$$f_{TE101} = \frac{1}{2\sqrt{\mu\epsilon}}\sqrt{\left(\frac{1}{0.04}\right)^2 + \left(\frac{1}{0.02}\right)^2} = 5\text{ GHz} \Rightarrow \epsilon_r = 2.81$$

$f_{TM110} = 3.73$ GHz $\Rightarrow$ single-mode operation at 3.73 – 5 GHz.

④ a) $\lambda = \frac{c}{f} = \frac{3\times10^8}{3\times10^6} = 100\ m$

$\frac{\ell}{\lambda} = \frac{1}{100} \Rightarrow$ Hertzian dipole

b) $E_\theta = \frac{I_0}{2r}\left(\frac{\ell}{\lambda}\right)\eta_0 = 5\times10^{-3}\ V/m$

$\Rightarrow I_0 = 2.65\ A$

$S_0 = 3.3\times10^{-8}\ W/m^2$

c) $P_{rad} = 0.277\ W$